

Compound AI System for Personalized Learning: Integrating LLM Agents with Knowledge Graphs

Eduard Antonov, M.Sc.

Master's Student, Applied Mathematics and Computer Science

Moscow Institute of Physics and Technology

Dolgoprudny, Moscow Oblast, 141700, Russia

Email: antonov.ek@phystech.edu

Abstract—Personalized learning plays a critical role in enhancing educational outcomes by customizing experiences to meet individual learners' unique needs, preferences, and abilities. Although Large Language Models (LLMs) offer promising opportunities for personalization through natural language interactions, they face challenges related to maintaining long-term context and ensuring traceability of a learner's evolving knowledge. This paper introduces a composite AI framework that merges an LLM-powered conversational agent with Knowledge Graphs (KGs) structured according to Knowledge Space Theory (KST), enhanced by advanced retrieval tools and a robust long-term memory mechanism. This system autonomously orchestrates meaningful, context-aware dialogues while utilizing function calling to access various tools, effectively bridging the gap between dynamic interactions and structured knowledge representation. Our empirical evaluations indicate that the proposed system excels in maintaining coherent dialogues, accurately tracking learner progress, and providing reliable, personalized educational support. By addressing the limitations of standalone LLMs and traditional adaptive learning systems, this integrated approach signifies a notable advancement in intelligent educational technology.

Keywords—personalized learning; large language models; ai agents; knowledge graphs; retrieval-augmented generation

I. INTRODUCTION

Personalized learning customizes education to meet individual needs, thereby enhancing both engagement and performance. Early adaptive systems, such as Skinner's teaching machines [1] and subsequent conversational interfaces [2][3], aimed to individualize instruction but often fell short due to the complexities of language, which limited their capacity for true personalization.

Advancements in knowledge tracing models, notably Bayesian Knowledge Tracing (BKT) [4] and Deep Knowledge Tracing (DKT) [5], have improved the prediction of learner mastery by effectively modeling the acquisition of procedural knowledge. Nonetheless, these models predominantly depend on passive assessments and standardized testing, which inadequately address open-ended tasks and the dynamics of real-time learning [6].

Generative AI, particularly Large Language Models [7], holds transformative potential due to their advanced capabilities in natural language understanding and reasoning [8][9]. LLMs can engage learners in natural and adaptive dialogues, offering explanations, feedback, and assessments that are tailored to

individual needs. However, they face challenges [10] such as maintaining long-term context [11], providing reliable source attribution [12], and ensuring interpretability, all of which can undermine trust within educational environments.

This study proposes an AI architecture that integrates an LLM-based agent [13] with a Knowledge Graph [14], retrieval tools [15][16], and a long-term memory mechanism [17]. This system aims to enhance adaptability, continuity, and traceability in personalized learning environments, effectively overcoming the limitations associated with standalone LLMs.

II. RELATED WORK

Advancements in Large Language Models, Knowledge Graphs, and AI agents have independently contributed to sophisticated knowledge representation and conversational abilities in educational technology. The integration of KGs with language models has shown potential in generating responses grounded in factual knowledge and facilitating knowledge tracing [18-20]. However, the specific application of agents combined with KGs for personalized learning remains relatively unexplored, as most existing adaptive learning systems rely on static frameworks or predefined rule sets.

A. Adaptive Learning Path Recommendation

Various approaches have been developed to optimize learning paths based on factors like time constraints, learner performance, and knowledge prerequisites. An adaptive learning path recommender utilizing auxiliary Learning Objects (LOs) was introduced to enhance the learning experience by recommending additional content when learners do not achieve expected scores [21]. Another model proposed a multidimensional KG framework for learning path recommendations, organizing LOs into hierarchical classes and establishing semantic relationships to generate customized learning paths tailored to different learning objectives and preferences [22]. Additionally, a personalized learning path recommendation model based on Long Short-Term Memory (LSTM) neural networks was developed to predict learning paths by capturing hidden patterns in learner data [23].

B. Knowledge Space Theory and Knowledge Graphs

Methodologies like Knowledge Space Theory offer valuable approaches for structuring educational content and tracking a learner's mastery [24][25]. Incorporating KST principles into KGs allows for modeling prerequisite hierarchies that guide students through a logical progression of

concepts. This structured representation ensures foundational knowledge is solidified before advancing to more complex topics, enhancing the effectiveness of personalized learning systems.

C. Integration of AI Agents and Knowledge Graphs

While previous studies have explored agentic systems and the integration of KGs with language models for generating contextually relevant dialogue [18][26], the use of AI agents in conjunction with KGs for personalized learning is still underexplored. Existing systems often depend on static rule sets or single-dimensional relationships, lacking the dynamic adaptability required for personalized education.

D. Comparative Analysis

Building on these insights, our proposed compound AI architecture uniquely integrates the interactive capabilities of LLM-based agents with the structured knowledge representation provided by KGs, augmented by a long-term memory mechanism. By leveraging function-calling mechanisms and advanced retrieval tools, our system facilitates more meaningful and context-aware dialogues. This approach bridges the gap between dynamic interactions and structured knowledge representation, providing personalized and traceable educational support over extended learning periods.

Unlike traditional adaptive systems focused on optimizing learning paths through time and score estimations or neural network-based predictions, our system combines these strategies with the conversational strengths of LLMs. This integration enables the system to maintain long-term context, accurately track learner progress, and adapt dynamically to individual needs, marking a significant advancement in intelligent educational technology.

III. METHOD

A. Task Definition

The primary goal of the proposed system is to provide educational content that is meticulously tailored to individual learners. This involves engaging learners in dynamic and interactive dialogues that enhance understanding while continuously assessing progress. Based on the learner's input and current knowledge state, the system aims to generate responses that are relevant, accurate, coherent, traceable, personalized, and educationally valuable.

B. System Architecture

The proposed system architecture, illustrated in Figure 1, consists of four core components: the AI Agent, the Knowledge Graph, Long-Term Memory, and Tools.

1) *AI Agent*: The AI Agent functions as the decision-making core of the system, orchestrating actions based on learner inputs and the current Knowledge State. It processes learner queries, invokes necessary functions through function-calling mechanisms [13], makes informed decisions by considering the Knowledge State [24] and established educational objectives, and generates personalized responses by leveraging the integrated LLM.

2) *The Knowledge Graph* is a fundamental component of the architecture, offering a structured and comprehensive

representation of the educational domain. Organized according to KST principles, it models concepts as nodes and their prerequisite relationships as directed edges. This arrangement facilitates prerequisite hierarchies, ensuring that foundational concepts are mastered before advancing to more complex topics, while also supporting knowledge tracking by monitoring and updating learner mastery levels across interconnected concepts. Figure 2 visualizes the Knowledge Graph structure and illustrates a potential Knowledge State.

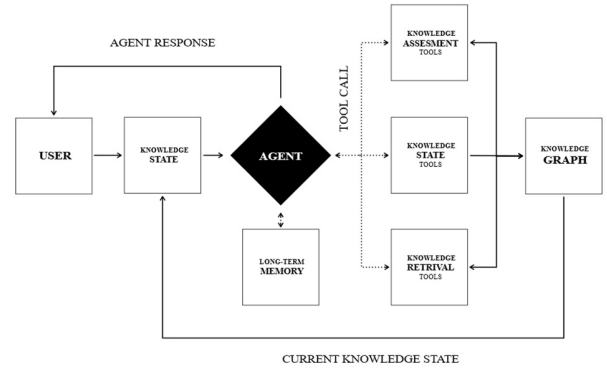


Fig. 1. Architecture of the system.

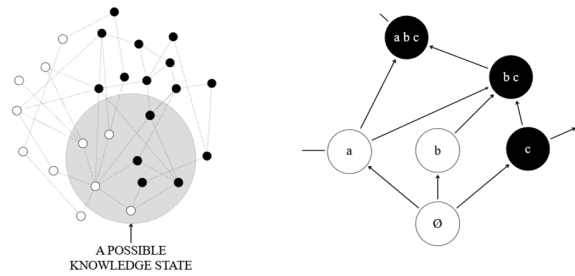


Fig. 2. The structure of the Knowledge Graph, presenting nodes as concepts and edges as prerequisite relationships. A shaded area indicates a potential Knowledge State, highlighting the subset of concepts a learner has currently mastered. Distinct sections or topic communities are represented by black and white nodes, enabling organized navigation and content delivery.

Formally, we define the Knowledge Graph as:

$$G = (N, E) \quad (1)$$

Where $N = \{c_1, c_2, \dots, c_n\}$ is the set of nodes (concepts), and $E \subseteq N \times N$ is the set of directed edges, with each edge $(c_i, c_j) \in E$ indicating that the concept c_i is a prerequisite for a concept c_j . This directed structure ensures learners follow logical progression paths through the material.

3) *The Knowledge State* (K_s) reflects the learner's current understanding and mastery within the subject domain. It is continuously updated based on ongoing interactions and performance metrics. Formally, the Knowledge State is represented as follows:

$$K_s = \{c_i: m_i | c_i \in N, m_i \in [0,1]\} \quad (2)$$

where c_i denotes a specific concept within the Knowledge Graph, and m_i represents the mastery level of c_i . This quantitative representation allows the system to precisely track and update the learner's progress across various concepts.

By ensuring that for any mastered concept $c_j \in K_s$, all prerequisite concepts c_i connected by directed edges $(c_i, c_j) \in E$ are also mastered, the graph helps guide the learner through a logical sequence of content.

4) *The Long-Term Memory* module is essential for maintaining continuity and personalization within learner interactions. It archives =historical interactions, learner preferences, and performance data, enabling the AI Agent to reference past conversations and adapt to the evolving needs of the learner. This functionality ensures that personalized learning experiences are sustained and progressively refined over time.

5) *The Tools* component includes auxiliary systems that support the AI Agent in fulfilling its functions. These encompass content retrieval mechanisms for accessing relevant sections of the Knowledge Graph based on learner queries and the current Knowledge State, evaluation modules that generate practice problems and assessments tailored to the learner's mastery levels and preferences, as well as data processing tools.

C. Knowledge Graph Construction and Expansion

To enhance scalability and minimize manual efforts in constructing the graph, we employ a hybrid approach that combines expert input with automated Natural Language Processing (NLP) techniques. Initially, subject matter experts and learning experience designers create a foundational KG by identifying core concepts and establishing prerequisite relationships based on educational standards and KST principles. This expert-constructed KG ensures that the essential structure and key educational objectives are accurately represented.

For automated expansion, we collect a corpus of domain-specific educational texts, such as glossaries, and reputable online resources. We apply NLP techniques to this corpus to extract additional concepts and relationships. After extracting these candidates, we validate the new concepts and relationships to ensure they are relevant and accurate. Validated concepts and relationships are then integrated into the existing KG, expanding it beyond the initial expert-defined structure.

Each concept in the Knowledge Graph is accompanied by embedded prompts designed to guide the Large Language Model in generating contextually relevant educational content. These dynamic prompts are generally universal but adapt to contextual needs each time a concept is retrieved. Examples include:

1) **Instructional Prompts:** Introduce concepts tailored to the learner's current understanding. For instance:

"Explain [concept] using a relatable scenario or analogy that highlights its real-world application. Use language and examples appropriate for the learner's current knowledge state."

This approach ensures that explanations and examples resonate with the learner, enhancing engagement and comprehension.

2) **Evaluation Prompts:** Generate targeted questions to assess the learner's grasp of specific concepts. For example:

"Create a multiple-choice question testing understanding of [concept], with one correct answer and three plausible distractors. Ensure the question focuses on distinguishing core aspects of [concept] from related but incorrect ideas."

These prompts are versatile and can be modified to optimize outcomes across topics, ensuring that the generated content aligns with educational objectives and maintains thematic consistency.

Combining expert oversight with automated processes, this approach supports efficient KG scaling while preserving accuracy and pedagogical integrity, enabling broad subject coverage without being limited by the manual efforts usually required in detailed knowledge graph construction.

D. Data Privacy and Security Measures

Sensitive user data, including content of the long-term module and personal information stored within the system, is encrypted to prevent unauthorized access, ensuring it remains strictly within the system's boundaries. Additionally, users have the option to review and manage stored data at any point during the session, offering transparency and control over what is retained.

E. Collaborative Learning and Multi-User Support

While the current system focuses on individual learner interactions, it also provides a foundation for collaborative learning. Future extensions could integrate a multi-agent framework where a supervisory agent manages individual agents for each learner. This setup would enable personalized engagement for each participant within a group learning environment without significant alterations to the core system.

IV. EXPERIMENTS

A. Implementation

To validate the proposed architecture, a prototype system was developed with a focus on introductory statistics—a foundational subject characterized by well-defined concepts and measurable mastery levels. The implementation details are as follows:

- **AI Agent:** Implemented using the ReAct [13] framework, orchestrates interactions by interleaving reasoning traces with task-specific actions, allowing it to adjust plans and handle exceptions dynamically. Through function-calling, it interacts with the Knowledge Graph and retrieval systems, ensuring accurate, reliable responses.
- **Language Model:** Utilized OpenAI's GPT-4, leveraging its advanced natural language processing capabilities to facilitate dynamic and coherent dialogues.

- **Knowledge Graph Construction:** Developed using Knowledge Space Theory principles, ensuring a structured and hierarchical representation of statistical concepts. Each node in the KG encapsulates specific attributes, including content descriptions and evaluation items.
- **Long-Term Memory:** Implemented using a vector database, enabling efficient storage and retrieval of interaction histories and learner data.
- **Custom Prompts:** Designed to guide the LLM in generating relevant educational content and assessments, ensuring alignment with the structured knowledge in the graph.

The prototype integrates these components seamlessly, providing a functional demonstration of the compound AI system's capabilities in a controlled educational setting.

B. Interaction Examples and Evaluation

The appendix provides detailed examples of interactions between the AI agent and learners, illustrating how the system leverages its components to create a personalized and adaptive learning experience. These examples demonstrate the agent's ability to maintain coherent dialogues, provide accurate explanations, and adapt to individual learner preferences.

C. Evaluation

To assess the effectiveness of the proposed system, we conducted preliminary evaluations focusing on both qualitative and quantitative aspects.

1) *Qualitative Evaluation:* Analyzing the interaction examples revealed several key strengths of the system. The AI agent maintained logical and consistent conversations, effectively preserving context over multiple interactions and enabling smooth transitions between topics. The system accurately reflected learner progress by continuously updating the Knowledge State, allowing for effective adjustment of content delivery based on evolving understanding.

The AI agent also adapted to learner's preferences and learning style, tailoring explanations and problem-solving approaches to improve engagement and comprehension. Responses were anchored in the Knowledge Graph, ensuring accuracy, transparency, and traceability to original concepts.

2) *Quantitative Evaluation:* To evaluate the system's performance from a technical standpoint, we measured several key operational metrics over a series of interactions.

- **Token Usage per Interaction (TUI):** The average number of tokens used in generating each response. Figure 3 depicts the Token Usage per Interaction (TUI) over a series of interactions.
- **Function-Call Accuracy (FCA):** The percentage of contextually correct tool invocations by the AI agent.
- **Response Time per Interaction (RTI):** The average time taken for the agent to generate a response.

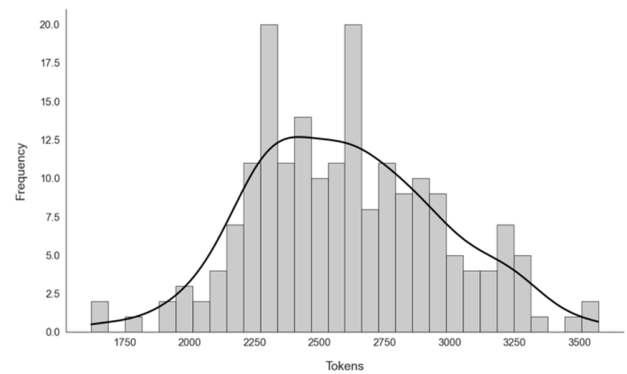


Fig. 3. Token Usage per Interaction (TUI)

In addition to token usage, the system demonstrated high Function-Call Accuracy and acceptable Response Time per Interaction. Specifically, the Function-Call Accuracy was measured at 94%, indicating that the AI agent reliably invoked the appropriate tools in contextually relevant scenarios. The Response Time per Interaction averaged 1.8 seconds, which is acceptable for real-time educational interactions.

By evaluating TUI, FCA, and RTI, we gain a comprehensive understanding of the system's operational effectiveness. Efficient token usage reflects the model's ability to generate concise and relevant responses, while high FCA ensures the integrity and reliability of interactions. Acceptable RTI contributes to a smooth and uninterrupted learning experience, which is vital for maintaining learner engagement and satisfaction.

D. Proposed Comprehensive Evaluation Plan

Evaluating agent-based AI systems for education presents unique challenges due to their complexity, real-world applications, and the need for nuanced, context-aware interactions [27]. Recognizing the need for a more thorough assessment, we propose a comprehensive evaluation plan to be implemented in future work:

- **User-Centered Testing:** Engaging a diverse group of learners to interact with the system in realistic educational settings. Surveys and interviews will capture feedback on accuracy, adaptability, personalization, continuity, and relevance to learning needs.
- **Expert Review:** Soliciting input from educational and AI experts to evaluate the system's instructional quality, alignment with pedagogical goals, and effectiveness in knowledge tracing and content delivery.
- **Performance Analysis:** Monitoring system metrics such as token usage, function-call accuracy, response time, and error rates to optimize operational efficiency and identify areas for improvement.

This multifaceted evaluation approach aims to comprehensively assess the system's ability to adapt dynamically, provide reliable support, and maintain continuity in learning interactions, with insights informing subsequent refinements.

V. DISCUSSION

The proposed compound AI system showcases significant potential in advancing personalized learning by dynamically adapting to individual learners and maintaining coherent, context-aware interactions. By integrating an LLM-based agent with a Knowledge Graph the system effectively addresses several key challenges in standalone LLM applications within educational contexts.

One of the system's primary strengths is its ability to provide context-aware dialogues that are both coherent and educationally valuable. By leveraging function-calling mechanisms, the AI agent can dynamically interact with various tools retrieve accurate information and tailor responses based on the learner's current knowledge state. This ensures that interactions are grounded in verified content, reducing the risk of misinformation—a common issue with generative models.

Empirical evaluations demonstrate the system's operational effectiveness. The AI agent reliably invokes appropriate tools and accesses relevant knowledge segments. Moreover, the hybrid approach to Knowledge Graph construction—combining expert input with automated Natural Language Processing techniques—facilitates scalability across diverse educational domains. This method allows the system to expand the Knowledge Graph without compromising accuracy or pedagogical integrity, enabling broader applicability.

Despite its strengths, the system faces certain limitations. Scalability remains a challenge when extending the KG to encompass more complex or varied subject areas. While the hybrid construction approach mitigates some scalability issues, further enhancements in automated KG expansion techniques are necessary to ensure seamless integration of new concepts and relationships without compromising quality.

Data privacy and security also present ongoing concerns. Although encryption measures are in place to protect sensitive user data, the long-term storage of learner interactions necessitates continuous monitoring and adherence to evolving privacy regulations. Implementing more sophisticated privacy-preserving techniques and ensuring compliance with international standards will be crucial as the system scales.

To comprehensively validate the system's effectiveness and adaptability, future evaluations should involve diverse learner populations across varied educational settings. Such evaluations will provide deeper insights into how the system performs in real-world scenarios and will highlight areas for refinement. Additionally, expanding the system to support collaborative learning environments through a multi-agent framework could further enhance its versatility, enabling personalized engagement for multiple learners simultaneously.

Moreover, integrating advanced analytics and feedback mechanisms could offer more nuanced insights into learner behaviors and preferences, allowing for even more tailored educational experiences. Exploring the incorporation of multimodal learning resources, such as visual aids and interactive simulations, could also enrich the learning process, catering to different learning styles and preferences.

VI. CONCLUSION

This paper introduced an innovative AI architecture that synergizes a Large Language Model-based agent with a Knowledge Graph structured by Knowledge Space Theory, supplemented by retrieval tools and a long-term memory module. The system effectively overcomes the limitations of standalone LLMs in personalized learning by maintaining coherent, context-rich dialogues and accurately tracking learner progress. Our preliminary evaluations demonstrate the system's ability to provide reliable and tailored educational support, marking a step forward in intelligent educational technology. Future work will focus on scaling the system to encompass broader subject areas, enhancing automated Knowledge Graph expansion techniques, and conducting comprehensive evaluations to further validate and refine the system.

APPENDIX

In this appendix, we provide detailed examples of interactions between the learner and the AI agent, along with the tools utilized by the agent in each scenario. These examples illustrate how the system leverages its components to create a personalized and adaptive learning experience.

Example 1: Greeting and Memory Recall

Learner: Hello, my name is Edward.

Agent: Hello, Edward! It's nice to meet you. How can I assist you with your learning today?

Tools used: Long-Term Memory

In this interaction, the agent stores the learner's name in the long-term memory to personalize future interactions.

Example 2: Topic Recommendation Based on Knowledge Graph

Learner: What would you recommend?

Agent: Since we're focusing on statistics, I can recommend starting with some foundational concepts. We have two main categories to explore:

1. Measures of Central Tendency: This includes concepts like mean, median, and mode, which are essential for understanding data sets.
2. Measures of Dispersion: This covers concepts such as range, variance, and standard deviation, which help in understanding the spread of data.

Would you like to start with an overview of one of these categories, or is there a specific concept you're curious about?

Tools Used: Section Retrieval from Knowledge Graph

The agent retrieves the main sections of the knowledge graph related to statistics to suggest foundational topics.

Example 3: Concept Explanation and Practice Suggestion

Learner: Let's begin with the median.

Agent: The median is a measure of central tendency that represents the middle value in a data set when the numbers are

arranged in order from least to greatest. It's particularly useful because it is less affected by outliers and skewed data than the mean.

Here's how you can find the median:

- For an Odd-Sized Data Set: Arrange the numbers in order and pick the middle one. Example: For the data set (3, 1, 4), arrange it as (1, 3, 4). The median is 3.
- For an Even-Sized Data Set: Arrange the numbers in order and take the average of the two middle numbers. Example: For the data set (3, 1, 4, 2), arrange it as (1, 2, 3, 4). The median is $(2 + 3) / 2 = 2.5$.

Would you like to try some practice problems to reinforce your understanding of the median?

Tools Used: Concept Retrieval by ID from Knowledge Graph

The agent retrieves the specific node for "median" from the knowledge graph and provides a structured explanation, including examples.

Example 4: Preference-Based Practice

Learner: Yes, and I prefer examples that incorporate astronomy and fun facts.

Agent: Let's dive into some practice problems with an astronomy twist and include some fun facts along the way!

Imagine you have the following distances of planets from the sun (in millions of kilometers): Mercury (57.9), Venus (108.2), Earth (149.6), Mars (227.9), and Jupiter (778.3).

To find the median distance, arrange these distances in order:

- Ordered distances: 57.9, 108.2, 149.6, 227.9, 778.3.

Fun Fact: Did you know that Earth is the only planet not named after a god?

Tools Used: Long-Term Memory, Evaluation Retrieval

The agent recalls the learner's preferences from long-term memory and retrieves an appropriate evaluation item, incorporating astronomy and fun facts.

Example 5: Answer Verification and Knowledge State Update

Learner: I think the answer is 149.6 million kilometers.

Agent: Great job, Edward! You've correctly identified the median distance. You've successfully completed this evaluation.

Would you like to try another problem or explore a different concept?

Tools Used: Knowledge State Update

The agent verifies the learner's answer, provides positive feedback, and updates the knowledge state in the knowledge graph to reflect the learner's mastery of the concept.

Example 6: Dynamic Progression Based on Graph Edges

Learner: Let's move on. Where would we go next?

Agent: Since you've mastered the concept of the median, a natural next step would be to explore the Interquartile Range (IQR). This concept measures the spread of the middle 50% of a data set, providing insight into data variability.

Would you like to learn more about the Interquartile Range, or is there another topic you're interested in?

Tools Used: Connections Retrieval from Knowledge Graph

The agent uses the knowledge graph's prerequisite structure to suggest the next logical concept, demonstrating dynamic adaptation based on the learner's progress.

REFERENCES

- [1] E. Watters, *Teaching Machines: The History of Personalized Learning*. Cambridge, MA: MIT Press, 2021, doi:10.7551/mitpress/12262.001.0001.
- [2] A. C. Graesser, K. VanLehn, C. P. Rose, P. W. Jordan, and D. Harter, "Intelligent Tutoring Systems with Conversational Dialogue," *AI Magazine*, vol. 22, no. 4, p. 39, 2001, doi:10.1609/aimag.v22i4.1591.
- [3] J. R. Carbonell, "AI in CAI: An Artificial-Intelligence Approach to Computer-Assisted Instruction," *IEEE Transactions on Man-Machine Systems*, vol. 11, no. 4, pp. 190-202, 1970, doi:10.1109/TMMS.1970.299942.
- [4] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253-278, 1994, doi:10.1007/BF01099821.
- [5] C. Piech et al., "Deep Knowledge Tracing," *arXiv preprint arXiv:1506.05908*, 2015. Available: <https://doi.org/10.48550/arXiv.1506.05908>.
- [6] S. Shen et al., "A Survey of Knowledge Tracing: Models, Variants, and Applications," *IEEE Transactions on Learning Technologies*, vol. 17, pp. 1898-1919, 2024, doi:10.1109/TLT.2024.3383325.
- [7] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, "Large Language Models: A Survey," *arXiv preprint arXiv:2402.06196*, 2024. Available: <https://doi.org/10.48550/arXiv.2402.06196>.
- [8] S. Bubeck et al., "Sparks of Artificial General Intelligence: Early experiments with GPT-4," *arXiv preprint arXiv:2303.12712*, 2023. Available: <https://arxiv.org/abs/2303.12712>.
- [9] Q. Niu et al., "Large Language Models and Cognitive Science: A Comprehensive Review of Similarities, Differences, and Challenges," *arXiv preprint arXiv:2409.02387*, 2024. Available: <https://arxiv.org/abs/2409.02387>.
- [10] Q. Li et al., "Adapting Large Language Models for Education: Foundational Capabilities, Potentials, and Challenges," *arXiv preprint arXiv:2401.08664*, 2024. Available: <https://doi.org/10.48550/arXiv.2401.08664>.
- [11] J. Xu, A. Szlam, and J. Weston, "Beyond Goldfish Memory: Long-Term Open-Domain Conversation," *arXiv preprint arXiv:2107.07567*, 2021. Available: <https://doi.org/10.48550/arXiv.2107.07567>.
- [12] Z. Ji et al., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, 2022, doi:10.1145/3571730.
- [13] S. Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models," *arXiv preprint arXiv:2202.03629*, 2022. Available: <https://doi.org/10.48550/arXiv.2202.03629>.
- [14] L. Ehrlinger and W. Wöb, "Towards a Definition of Knowledge Graphs," in *SEMANTiCS (Posters, Demos, SuCCESS)*, vol. 48, pp. 1-4, 2016.
- [15] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *NeurIPS 2020*, 2020. Available: <https://doi.org/10.48550/arXiv.2005.11401>.

- [16] Y. Zhu et al., "Large Language Models for Information Retrieval: A Survey," *arXiv preprint arXiv:2308.07107*, 2023. Available: <https://doi.org/10.48550/arXiv.2308.07107>.
- [17] C. Packer et al., "MemGPT: Towards LLMs as Operating Systems," *arXiv preprint arXiv:2310.08560*, 2023. Available: <https://doi.org/10.48550/arXiv.2310.08560>.
- [18] M. Kang, J. M. Kwak, J. Baek, and S. J. Hwang, "Knowledge Graph-Augmented Language Models for Knowledge-Grounded Dialogue Generation," *arXiv preprint arXiv:2305.18846*, 2023. Available: <https://doi.org/10.48550/arXiv.2305.18846>.
- [19] H. Abu-Rasheed, C. Weber, and M. Fathi, "Knowledge Graphs as Context Sources for LLM-Based Explanations of Learning Recommendations," *arXiv preprint arXiv:2403.03008*, 2024. Available: <https://doi.org/10.48550/arXiv.2403.03008>.
- [20] U. Lee et al., "Language Model Can Do Knowledge Tracing: Simple but Effective Method to Integrate Language Model and Knowledge Tracing Task," *arXiv preprint arXiv:2406.02893*, 2024. Available: <https://doi.org/10.48550/arXiv.2406.02893>.
- [21] A. H. Nabizadeh, D. Gonçalves, S. Gama, J. Jorge, and H. N. Rafsanjani, "Adaptive learning path recommender approach using auxiliary learning objects," *Computers & Education*, 2019, Article ID 103777, doi:10.1016/j.compedu.2019.103777.
- [22] D. Shi, T. Wang, H. Xing, and H. Xu, "A learning path recommendation model based on a multidimensional knowledge graph framework for e-learning," *Knowledge-Based Systems*, 2020, Article ID 105618, doi:10.1016/j.knosys.2020.105618.
- [23] Y. Zhou, C. Huang, Q. Hu, J. Zhu, and Y. Tang, "Personalized learning full-path recommendation model based on LSTM neural networks," *Information Sciences*, vol. 423-424, pp. 53-65, 2018, doi:10.1016/j.ins.2018.02.053.
- [24] J.-P. Doignon and J.-C. Falmagne, "Spaces for the Assessment of Knowledge," *International Journal of Man-Machine Studies*, vol. 23, pp. 175-196, 1985.
- [25] J.-P. Doignon and J.-C. Falmagne, "Knowledge Spaces and Learning Spaces," *arXiv preprint arXiv:1511.06757*, 2015. Available: <https://doi.org/10.48550/arXiv.1511.06757>.
- [26] H. Li, C. Yang, A. Zhang, Y. Deng, X. Wang, and T.-S. Chua, "Hello Again! LLM-powered Personalized Agent for Long-term Dialogue," *arXiv preprint arXiv:2406.05925*, 2024. Available: <https://doi.org/10.48550/arXiv.2406.05925>.
- [27] S. Kapoor, B. Stroebel, Z. S. Siegel, N. Nadgir, and A. Narayanan, "AI Agents That Matter," *arXiv preprint arXiv:2407.01502*, 2024. Available: <https://doi.org/10.48550/arXiv.2407.01502>.